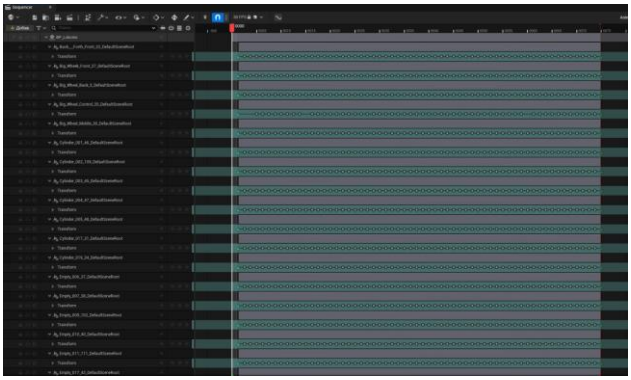
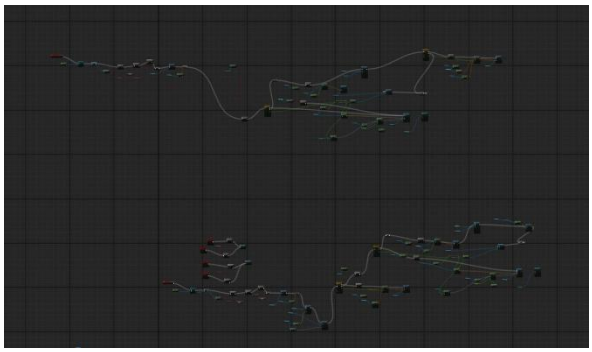
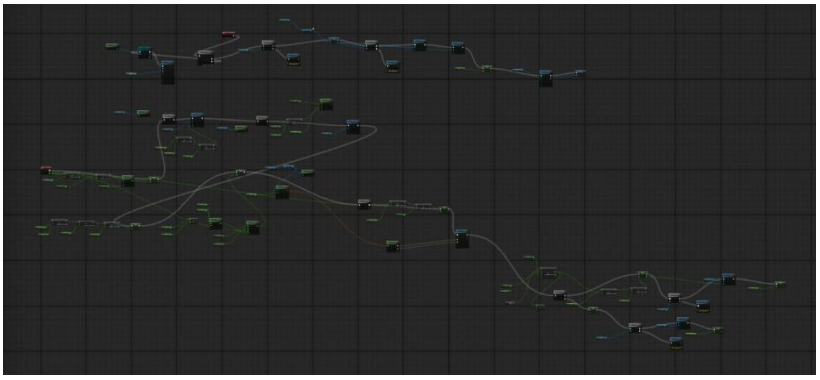
Report 2
TRAINDRAIN

Team 08: Volodymyr Zaikin, Aleksandr Zhidkov



(Engine for train)

(Train blueprint)

Current status

Game design
TrainDrain is still focused on driving the train as far as possible while keeping it from being destroyed by monsters, but we have advanced the internal mechanics of the train itself.
• The complete concept for train and wagon durability is now defined: monsters and furnace overheating reduce durability, and specific parts of the train can enter “damaged” and “critical” states.
• A detailed repair system for the locomotive engine has been designed: the player will interact with engine components inside the cabin to restore durability.
• The relationship between three core actions (throwing coal, shooting monsters, repairing the train) has been clarified so that the player must constantly prioritize between them.
• Conditions for train destruction and run failure are finalized based on the new durability and repair system.

Programming part
• A technical design for the destruction and repair mechanics has been prepared, including variables for durability, damage thresholds and repair actions to be used in Blueprints.
• The logic of how damage from monsters and furnace overheating will be applied to the train is

described and broken down into separate events (hit events, overheat ticks, repair actions).
• Dependencies between train speed, furnace state and train durability are outlined for later integration into the existing train movement system.

• Blueprint now includes synchronization of wheel rotation with the train's current speed, so that the visual animation fully corresponds to the actual movement.

Graphics part
• Work on weapon models has started: the first pistol based on real-life references is being modeled and adjusted for first-person use.
• The existing train model is being modified in Blender to include a visible engine area inside the locomotive where the player will perform repairs.
• Additional details are being added to the locomotive interior so that damage/repair zones are visually readable (panels, engine parts, access hatch).

Plans for the near future

Game design
• Fine-tune the repair mechanics (repair speed, resource cost, cooldowns) to keep the balance between shooting, feeding coal and fixing the train.
• Define different levels of malfunctions for the engine (minor damage vs. critical failure) that will require different amounts of repair time.
• Adjust difficulty progression so that train damage, monster pressure and furnace management scale together over time.

Programming part
• Implement the designed train damage, destruction and repair system inside the train Blueprint in Unreal Engine.
• Integrate the new durability system with existing mechanics: train movement, furnace overheating and monster attacks.
• Add simple feedback for the player (e.g. basic UI indicators or on-screen messages) to show train condition and repair progress.
• Refactor the train Blueprint structure to make future additions (extra wagons, new damage types) easier to implement.

Graphics part
• Finish the pistol model, add UVs and basic textures, and export it to Unreal Engine to replace the placeholder gun.
• Complete the updated locomotive model with the detailed engine compartment and ensure that all repair interaction points are accessible and readable in first person.
• Prepare damaged visual states for engine parts (bent panels, smoke, sparks placeholders) to support the destruction and repair system.

Time allocation

- Volodymyr Zaikin total 9 hours
- Aleksandr Zhidkov total 9 hours